

Question 1 (12 marks)**Marks**

- a) Solve: $2\sin x = \sqrt{3}$ $0^\circ \leq x \leq 360^\circ$ **2**
- b) Express $0.424242\dots$ as a fraction (showing working) in the form $\frac{p}{q}$, where p and q are integers. **2**
- c) Solve for x : i) $|x + 7| = 3x - 2$ **3**
- ii) $2^{5x-2} = 32$ **2**
- d) If $(5 - 3\sqrt{2})(4 + \sqrt{2}) = a + b\sqrt{c}$, find the values of a , b and c . **3**

Question 2 (12 marks)

- a) Evaluate i) $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$ **1**
- ii) $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x + 1}{5 - 4x - 2x^2}$ **1**
- b) Find $\frac{dy}{dx}$, given that **7**
- i) $y = \sqrt{x^3}$
- ii) $y = 3x^3 - 7x + 6$
- iii) $y = 7x(4x + 8)^6$
- iv) $y = \frac{x - x^2}{5x + 1}$
- c) Differentiate from First Principles $y = x^2 - 3x$ **3**

Question 3 (12 marks)

- a) Sketch the following functions on separate number planes. Identify any intercepts and asymptotes where appropriate. **6**

i) $y = \frac{-1}{x+3}$

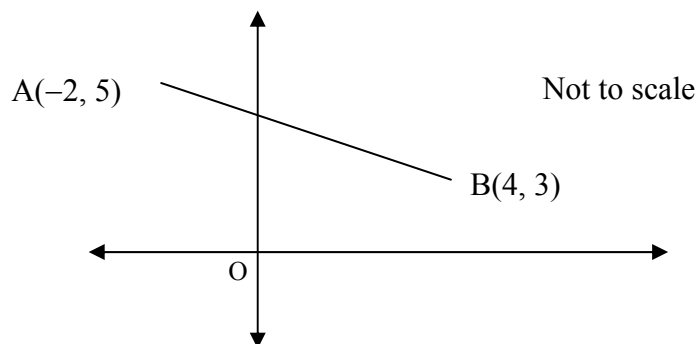
ii) $y = \sqrt{5-x}$

- b) Solve: $2\log_a(x-4) - \log_a(x-5) = \log_a(x-2)$ **3**

- c) Solve: $\frac{5x}{2x-1} \geq 3$ **3**

Question 4 (12 marks)

- a)



The diagram shows the points A(-2, 5), B(4, 3) and O(0, 0).
The point C is the fourth vertex of the parallelogram OABC.

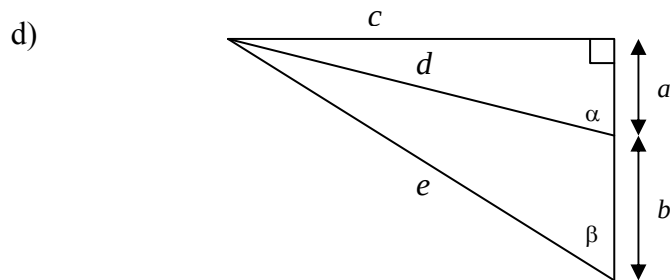
- i) Show by derivation that the equation of AB is $x + 3y - 13 = 0$ **1**
- ii) Find in exact form the length of AB. **1**
- iii) Calculate the perpendicular distance from O to AB. **2**
- iv) Calculate the area of parallelogram OABC. **1**
- v) Find the coordinates of C. **2**
- b) Sketch the region represented by $x^2 + (y-1)^2 \leq 9$ **2**
- c) Find the equation of the line through the point of intersection of the lines $2x - 3y + 6 = 0$ and $5x + y - 4 = 0$ and the point (1, 4). **3**

Question 5 (12 marks)

a) Sketch the curve $y = 3\cos 2x$ $0^\circ \leq x \leq 360^\circ$ 3

b) A, B and C are three towns. B is 20km from A in the direction 330°T .
C is 30km from A in the direction 205°T . Find the distance from B to C.
(Hint: draw a diagram) 3

c) Prove the identity: $\frac{1}{\sec \theta - \tan \theta} - \frac{1}{\sec \theta + \tan \theta} = 2 \tan \theta$ 3



Show that $a = \frac{b \cos \alpha \sin \beta}{\sin(\alpha - \beta)}$ 3

Question 6 (12 marks)

a) Find the sum of the first 20 multiples of 7. i.e. $7 + 14 + 21 + \dots$ 2

b) A gardener weeded his lawn over summer. The first year he dug out 2 wheelbarrows full. Each successive year he dug out $\frac{3}{4}$ of the previous years total.

i) How many barrows full will he have dug out in the first 10 years? 2

ii) Over his lifetime, what is the limiting number of barrow loads he will end up removing? 2

c) If $2p + 1$, $5p$, $12p - 4$ are the first 3 terms of a geometric sequence, find the value of p and hence find T_n . 3

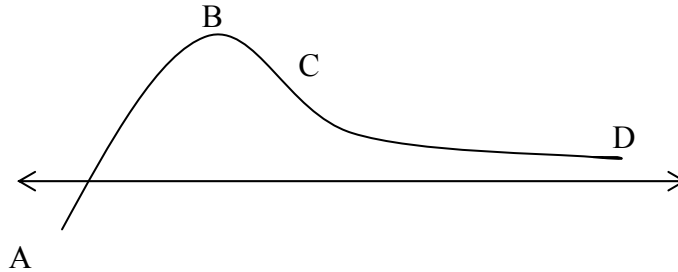
d) Consider the series $\log_a 36 + \log_a 18 + \log_a 9 + \dots + \log_a \frac{9}{8}$ 3

i) Show that it is an arithmetic series.

ii) Find the sum of the series.

Question 7 (12 marks)

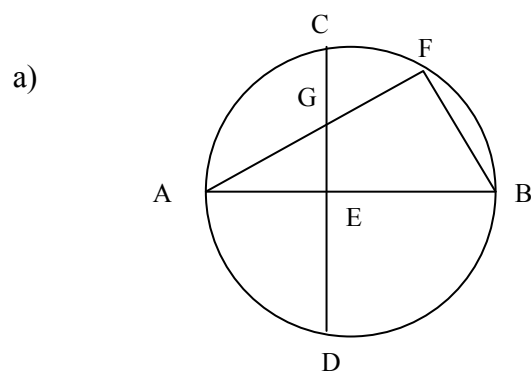
- a) Sketch the derivative of the curve below and clearly label it 2



- b) Find the equation of the normal to $y = 9 - x^2$ at $P(1, 8)$. 3

- c) Given the curve $y = x^4 - 16x^3 + 72x^2 + 10$, find 7

- i) All stationary points and determine their nature
- ii) Any points of inflexion, justifying your answers
- iii) Sketch the curve showing this information.

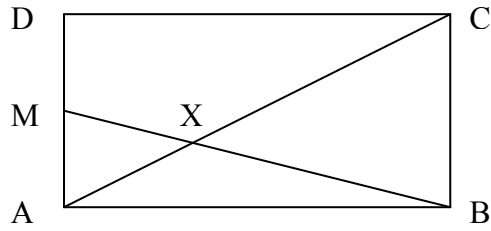
Question 8 (12 marks)

AB is a diameter and $CD \perp AB$.

Prove that E, G, F and B are concyclic

3

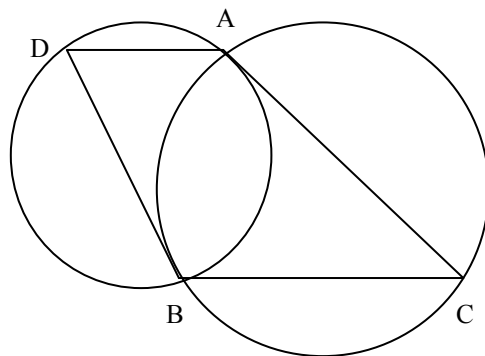
b)



ABCD is a rectangle. $AB = 2AD$. M is the midpoint of AD. The line BM meets AC at X.

- i) Show that $\triangle AXM$ and $\triangle BXC$ are similar. 3
- ii) Show that $3CX = 2AC$ 2

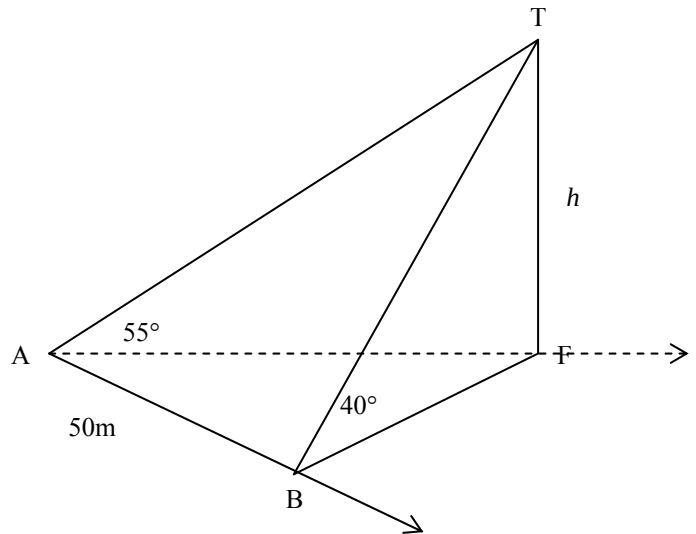
- c) Two circles meet each other at A and B. AC and DB are tangents. Prove that $AD \parallel BC$ 4



Question 9 (12 marks)

a)

The diagram shows a tower TF of height h metres standing due north of A on level ground. The angles of elevation of the top T of the tower from two points A and B (due east of A), on the ground nearby are 55° and 40° respectively. The distance AB is 50m.



- | | | |
|------|--|---|
| i) | What is the measure of $\angle FAB$? | 1 |
| ii) | Find AF and BF in terms of h . | 2 |
| iii) | Hence find the height of the tower to the nearest metre. | 3 |

b) A cylinder, open at one end has a volume of 1000cm^3 .

- | | | |
|-----|---|---|
| i) | Show that the surface area S is given by $S = \pi r^2 + \frac{2000}{r}$ | 2 |
| ii) | Find the value of r , to 4 significant figures that minimises the surface area. | 4 |

Question 10 (12 marks)

- a) A ball is thrown vertically up in the air with its height x metres above the ground at any time t seconds given by $x = 4t(5 - t)$.
- | | | |
|------|--|---|
| i) | When does it reach maximum height? | 2 |
| ii) | What is the maximum height reached? | 1 |
| iii) | What is its acceleration then? | 1 |
| iv) | What is the speed of the ball when it returns to the ground? | 2 |
| v) | Find the distance travelled during the 3 rd second. | 2 |
- b) Sketch a continuous curve $y = f(x)$ having the following properties.
- $f(-3) = 12,$
 $f(0) = 6,$
 $f(3) = f'(3) = f'(-3) = 0$
 $f'(x) < 0$ for $-3 < x < 3$ and $f'(x) > 0$ for $x < -3$ or $x > 3$
- c) Give an example by sketching, of a function which has a minimum at $x = 0$, but which is not differentiable at $x = 0$.

END OF EXAMINATION

2008 YR11 MATHS SUGGESTED SOLUTIONS

Q1

$$2 \sin x = \sqrt{3}$$

$$\sin x = \frac{\sqrt{3}}{2}$$

$$x = 60^\circ \text{ or } 120^\circ$$

1, 1

b) $x = 0.4242 \dots$

$$100x = 42.42 \dots$$

$$99x = 42$$

$$x = \frac{42}{99}$$

$$= \frac{14}{33}$$

1

c) i) $|x+7| = 3x-2$

$$x+7 = 3x-2$$

$$9 = 2x$$

$$x = 4\frac{1}{2}$$

$$2^{5x-2} = 32 \Rightarrow 5x-2 = 5 \quad 2.$$

$$-x-7 = 3x-2 \quad x = \frac{7}{5}$$

$$4x = -5$$

$$x = -\frac{5}{4}$$

1

Test Solns

$$4\frac{1}{2} \sqrt{-\frac{5}{4}} \times$$

only one

$$x = 4\frac{1}{2}$$

$$(4+\sqrt{2})(4+\sqrt{2}) = 20+5\sqrt{2}-12\sqrt{2}-6$$

$$14-7\sqrt{2}$$

$$a=14$$

$$b=-7$$

$$c=0$$

Q2

a) i) $\lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{(x-4)} = 8$

1

ii) $\frac{3}{-2}$

1

b) i) $y = x^{3/2}$ Accept $\frac{3x^{1/2}}{2\sqrt{x^3}}$

$$y' = \frac{3}{2} x^{1/2}$$

1

ii) $y' = 9x^2 - 7$
transcript error $6x-7$ (Award 2)

2

iii) $y = 7x(4x+8)^6$ product rule otherwise 0.

$$y' = 7x \times 6 \times 4(4x+8)^5 + (4x+8)^6 \times 7$$

1

$$= 168(4x+8)^5 + 7(4x+8)^6$$

$$= 7(4x+8)^5 [24 + 7(4x+8)]$$

mistake after not limited

iv) $y = \frac{x-x^2}{5x+1}$ wrong rule = 0.!!

$$y' = \frac{(5x+1)(1-2x) - (x-x^2) \times 5}{(5x+1)^2}$$

1, 1

$$= \frac{-5x^2 - 2x + 1}{(5x+1)^2}$$

has to be right

e) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h}$$

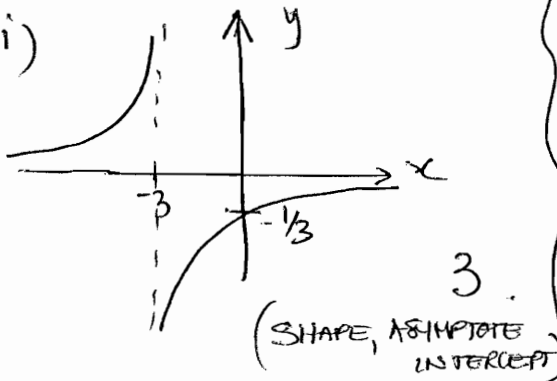
$$= \lim_{h \rightarrow 0} \frac{2xh - 3h + h^2}{h}$$

$$= 2x - 3$$

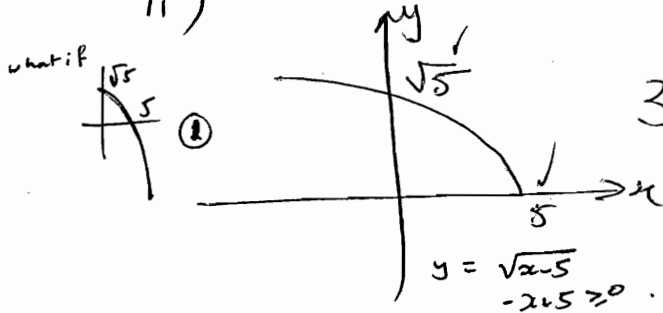
lose a mark for no $\lim_{h \rightarrow 0}$

Q3

a) i)



ii)



b) $2\log(x-4) - \log(x-1) = \log(x)$

$$\frac{(x-4)^2}{x-1} = x+2$$

$$(x-4)^2 = (x-1)(x+2)$$

$$x^2 - 8x + 16 = x^2 - 7x + 2$$

$$x = 6$$

Ignoring denominator = 0

$$5x \geq 3 \quad (2x-1)(-x+3) \geq 0$$

$$\frac{2x-1}{-x+3} \geq 0$$

$$x > \frac{1}{2}$$

case 2 $\frac{2x-1}{-x+3} < 0$

$$x < \frac{1}{2}$$

$$5x \geq 6x-3$$

$$x \geq -3$$

$$x \leq 3$$

$$5x \leq 6x-3$$

$$-x \leq -3$$

$$x \geq 3$$

No soln

Only soln

$$\frac{1}{2} < x \leq 3$$

Q4

a) i) $m_{AB} = \frac{5-3}{-2-4} = -\frac{1}{3}$

$$4-5 = -\frac{1}{3}(x+2)$$

$$3y-15 = -x-2$$

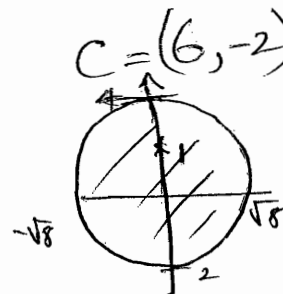
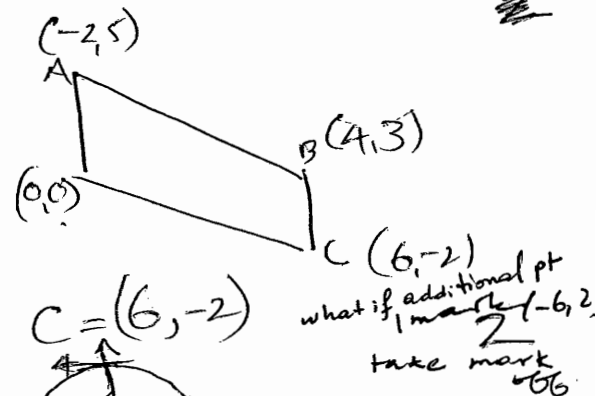
$$x+3y-13=0$$

ii) $AB = \sqrt{(-2-4)^2 + (5-3)^2}$
 $= \sqrt{40}$
 $= 2\sqrt{10}$

iii) $d = \frac{|0 + 3 \times 0 - 13|}{\sqrt{1+9}}$
 $= \frac{13}{\sqrt{10}} = \frac{13}{\sqrt{10}}$

iv) $A = bh = 2\sqrt{10} \times \frac{13}{\sqrt{10}} = 26u^2$

v)



1 circle

1 region

Note centre necessary

maybe 4 or 2
sqrt Not necessary

c) $(2x-3y+6) + k(5x+y-4) = 0$
Subst $(2-12+6) + k(5+4-4) = 0$
 $5k = 4$
 $k = 4/5$

$$\Rightarrow 5(2x-3y+6) + 4(5x+y-4) = 0$$

Q4-

$$\begin{aligned} \text{c) } 2x - 3y + 6 &= 0 \\ 5x + y - 4 &= 0. \times 3 \end{aligned}$$

$$\begin{aligned} 2x - 3y &= -6 \\ 15x + 3y &= 12 \\ \hline 17x &= 6. \\ x &= 6/17. \end{aligned}$$

$$2\left(\frac{6}{17}\right) - 3y + 6 = 0.$$

$$12 - 3y + 12 = 0.$$

$$y = \frac{114}{51} = 2\frac{4}{17}.$$

$$\left(\frac{6}{17}, 2\frac{4}{17}\right) \checkmark$$

(1, 4)

$$m = \frac{4 - 2\frac{4}{17}}{1 - 6/17} = 2\frac{8}{11} = \frac{30}{11}. \checkmark$$

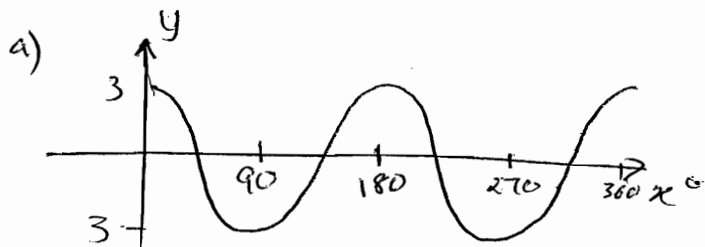
$$y - 4 = \frac{30}{11} (x - 1)$$

$$11y - 44 = 30x - 30.$$

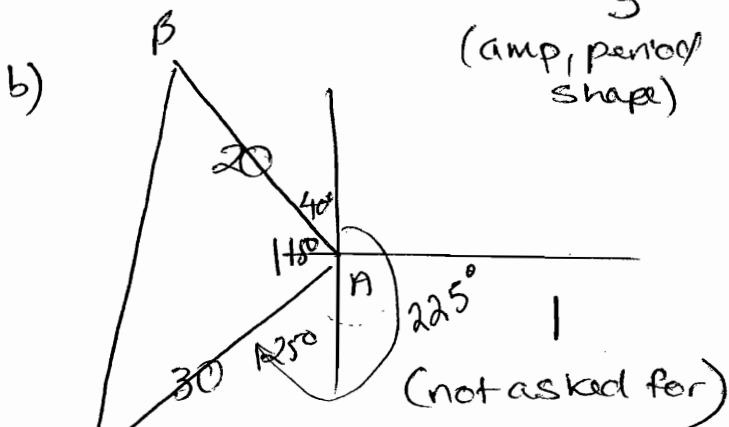
$$0 = 30x - 11y + 14$$

$$\rightarrow y = \frac{30x}{11} + \frac{14}{11} \checkmark$$

Q5



3
(amp, period
shape)



(not asked for)

$$BC^2 = 20^2 + 30^2 - 2 \times 20 \times 30 \times \cos 125^\circ$$

$$= 1867.142$$

$$BC \div 42.5 \text{ km } 44.6 \text{ km}$$

(Note: 1P no diagram
2 marks at point 2)

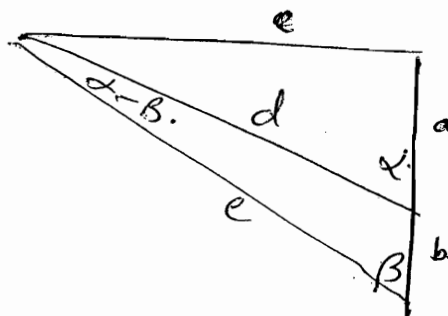
c) LHS = $\frac{1}{\sec \theta + \tan \theta} - \frac{1}{\sec \theta + \tan \theta}$

$$= \frac{\sec \theta + \tan \theta - \sec \theta + \tan \theta}{\sec^2 \theta - \tan^2 \theta}$$

$$= \frac{2 \tan \theta}{1}$$

$$= \text{RHS}$$

d)



$$\cos \alpha = \frac{a}{d}$$

$$d = \frac{a}{\cos \alpha}$$

$$\frac{b}{\sin(\alpha - \beta)} = \frac{d}{\sin \beta}$$

$$\frac{b}{\sin(\alpha - \beta)} = \frac{a}{\cos \alpha \sin \beta}$$

$$a = \frac{b \cos \alpha \sin \beta}{\sin(\alpha - \beta)}$$

$$\sin \alpha \cos \beta - \cos \alpha \sin \beta$$

Q6

a) $a=7$

$l=140$

$n=20$

$$S = \frac{20}{2} (7 + 140)$$

$= 1470$

b) $r = \frac{3}{4}$ $a=2$ $n=10$

$$S_{10} = 2 \left(\frac{1 - \left(\frac{3}{4}\right)^{10}}{1 - \frac{3}{4}} \right)$$

$\div 7.5 \text{ loads}$

7 loads (2)

c) $S = \frac{a}{1-r} = \frac{2}{1-\frac{3}{4}} = 8 \text{ loads}$

d) $\frac{5p}{2p+1} = \frac{12p-4}{5p}$

$25p^2 = 24p^2 + 4p - 4$

$(p-2)^2 = 0$

$p=2$

$T_2 = 10$ $T_3 = 20$
asked for

$5 \times 2^{n-1}$

d)

i) $T_3 - T_2 = \log 9 - \log 8$
 $= \log \frac{1}{2} = -\log 2$

$T_2 - T_1 = \log 18 - \log 36$
 $= \log \frac{1}{2} = -\log 2$
 $= -\text{absol.}$

$\therefore$ arithmetic.

+ log 2 no marks.

ii) $n=6$

$a = \log 36$

$l = \log \frac{9}{8}$

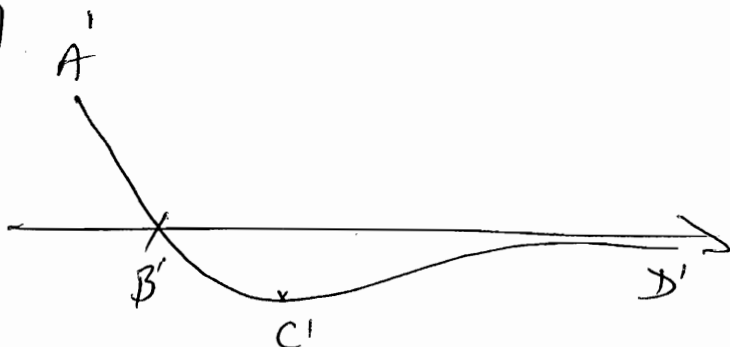
$$S = \frac{6}{2} \left[\log 36 + \log \frac{9}{8} \right]$$

$= 3 \log \left(\frac{81}{2} \right)$

$\frac{9}{26} \times \frac{9}{8}$

Q7

a)



2

b) $y = 9 - x^2$

$y' = -2x$

at $x = 1$, $m = -2 \therefore \perp m = \frac{1}{2}$

$y - 8 = \frac{1}{2}(x - 1)$

$2y - 16 = x - 1$

$x - 2y + 15 = 0$

for tangent: (2) marks
if correct

c) $y^4 - 16x^3 + 72x^2 + 16$

$y' = 4x^3 - 48x^2 + 144x$

$= 4x(x^2 - 12x + 36)$

$= 4x(x - 6)^2$

$y'' = 12x^2 - 96x + 144$

$= 12(x - 2)(x - 6)$

$y' = 0 \Rightarrow x = 6 \text{ or } 0$

$y = 432$

$(6, 432)$

$(0, 16)$

$y'' = 0 \Rightarrow x = 2 \text{ or } 6$

$(2, 176)$
186

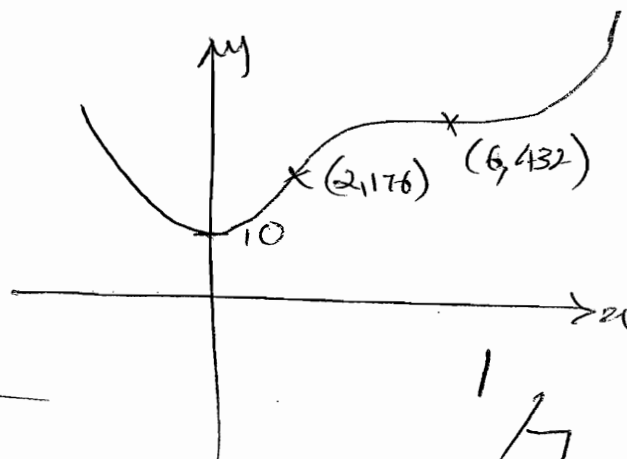
$(6, 432)$
442
7354

Test

x	-1	0	1	6	8
y'	<0	0	>0	0	>0

 $\therefore$ Min at $x = 0$ horiz inflex at $x = 6$

x	0	2	4	6	8
y''	>0	0	<0	0	>0



Q8

a) $\angle AFB = 90^\circ$ (diameter/circ. thm) |

$\angle CEB = 90^\circ$ (given) |

$\therefore ECFB$ is a cyclic Quad. (Opp $\angle$ s Σ to 180°) |

b) i) $\angle MXA = \angle CXB$ (Vert opp) |

$\angle DAX = \angle ACB$ (alts =, $DC \parallel AB$) |

$\therefore \triangle MXA \parallel \triangle CXB$ (AA) |

ii) $2MA = CB$ (given)

$2AX = XC$ (Corrs sides Sim Δ s)

$AX + CX = AC$
 $3AX = AC$
 $\therefore \frac{CX}{AX} = 2$

$CX = \frac{2}{3}AC$

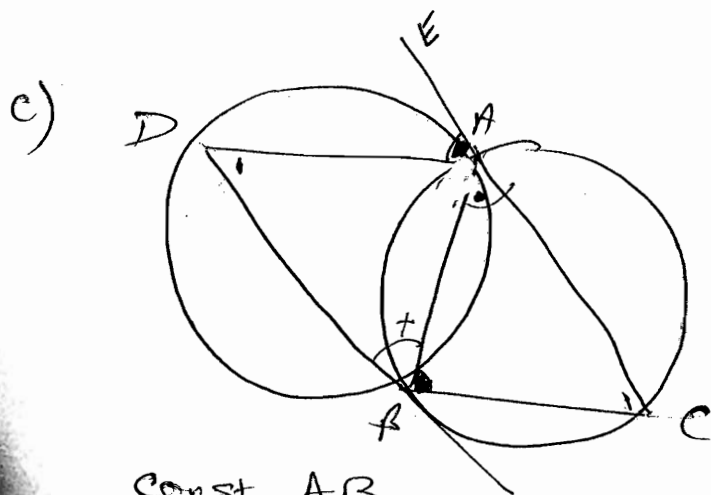
$3CX = 2AC$

h.w?

$CX = 2AX$
 $3CX = 6AX$

$AC = 3AX$
 $2AC = 6AX$

$6AX = 3CX = 2AC$ |



$\angle DAC + \angle ACB = 180^\circ$
 $\therefore DA \parallel BC$ (co-interior)
 gets 0 marks!

Const AB

let $\angle BCA = \alpha$

$\therefore \angle ABD = \alpha$ (tangent alt seg thm) |

let $\angle BDA = \beta$

$\therefore \angle CAB = \beta$ (" " " ") |

$\therefore \angle ABC = 180 - (\alpha + \beta)$ ($\angle$ Sum Δ) |

$\angle DAB = 180 - (\alpha + \beta)$ (" ") |

$\therefore \angle ABC = \angle DAB$ |

Q9

a) i) $\angle FAB = 90^\circ$

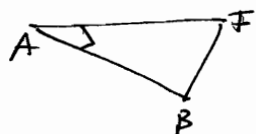
ii) $\frac{AF}{h} = \cot 55^\circ$

$BF = h \cot 40^\circ$

h in terms of m

$AF = h \cot 55^\circ$

iii) $(h \cot 55^\circ)^2 + 2500 = (h \cot 40^\circ)^2$



$h^2 \cot^2 55^\circ + 2500 = h^2 \cot^2 40^\circ$

$h^2 = \frac{2500}{\cot^2 40^\circ - \cot^2 55^\circ} = \frac{2500}{\tan^2 50^\circ - \tan^2 35^\circ}$

≈ 2688.2

$h \approx 51.8 \text{ m} \approx 52 \text{ m}$

b) i) $V = \pi r^2 h \Rightarrow 1000 = \pi r^2 h$

$h = \frac{1000}{\pi r^2}$

$SA = \pi r^2 + 2\pi r h$

$= \pi r^2 + \frac{2000}{r}$

ii)

$\frac{d(SA)}{dr} = 2\pi r - \frac{2000}{r^2}$

$\frac{dSA}{dr} = 0 \Rightarrow 2\pi r = \frac{2000}{r^2}$

$r^3 = \frac{1000}{\pi}$

$r = \frac{10}{\sqrt[3]{\pi}}$

Test

r	$\frac{10}{\sqrt[3]{\pi}} -$	$\frac{10}{\sqrt[3]{\pi}}$	$\frac{10}{\sqrt[3]{\pi}} +$
SA	-	0	+

$\therefore \text{Min } r \approx 6.8278 = \sqrt[3]{\frac{1000}{\pi}}$

4 sig figs

6.828

Q 10

a) i) $x = 20t - 4t^2$

$\dot{x} = 20 - 8t$

(1) or $t = \frac{-b}{2a} = \frac{-20}{-8} = 2\frac{1}{2}$

$\dot{x} = 0 \Rightarrow t = 2\frac{1}{2}$ (1)

ii) $x(2\frac{1}{2}) = 20 \times 2\frac{1}{2} - 4 \times (2\frac{1}{2})^2 = 25m.$
 ~~$39\frac{3}{4}$~~

iii) $\ddot{x} = -8 \text{ m/s}^2$

iv) at $t = 5$

$\dot{x}(5) = 20 - 8 \times 5 = -20 \text{ m/s}$
 speed = 20 m/s .

v) $x(2) = 24$ $x(2\frac{1}{2}) = 25$ $x(3) = 24$

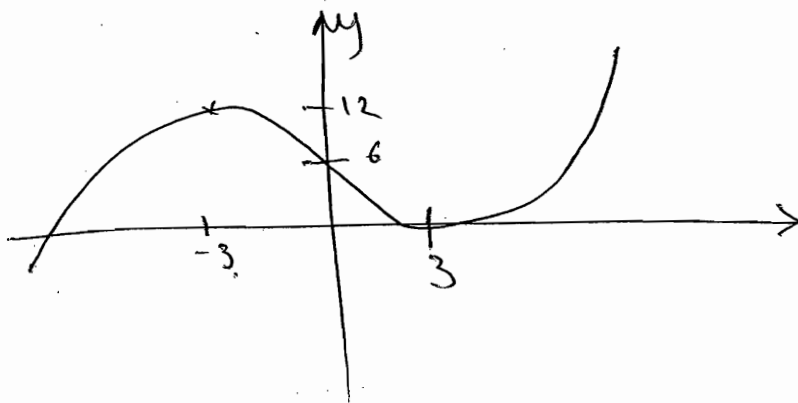
~~Limit of 5.~~

$\therefore \text{distance} = 3\frac{1}{2}^2 \text{ m.}$

3 $\rightarrow$ 4 :

2

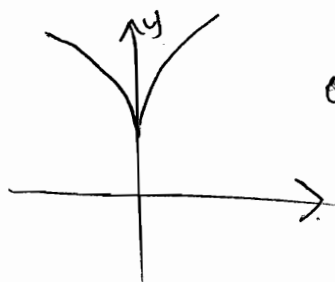
b)



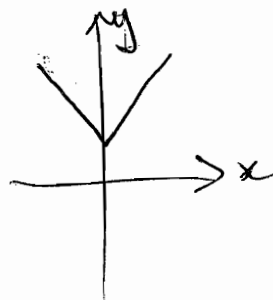
3
(Max, Min, Inflex)

c)

eg



or



Accept
Accept

or anything else that
meets the criteria